



Module: Order Management System API Documentation

Group-2

Presented by:
2105054 (Md. Rafat Kabir)

Other Members of the group:
2105037(Anuron Maitro)
2105038(Md. Asif Ali)
2105042(Md. Maruf Hasan Rubab)
2105047(Himadri Gobinda Biswas)
2105052(Md. Mehedi Hasan)

HTTP Status Codes

Code	Description
201	Successful Outcome
400	Bad request - validation failed
401	Unauthorized - missing or invalid authentication
500	Internal server error

SECTIONS

- USER AUTHENTICATION & PROFILE
- PRODUCTS
- CART
- ORDERS
- SHIPPING ADDRESS MANAGEMENT
- PRODUCT REVIEWS & RATINGS
- USER WISHLIST MANAGEMENT
- PAYMENT INTEGRATION

USER AUTHENTICATION & PROFILE

Authentication Endpoints

POST

`/api/auth/sync`

Sync user from Clerk

PUBLIC

GET

`/api/users/profile`

Get user profile

AUTH

PUT

`/api/user/profile`

Update user profile

AUTH

Authentication uses Clerk JWT token passed in Authorization: Bearer {token} header

User Authentication & Profile

Sync user from Clerk

POST

/api/auth/sync

PUBLIC

Registers or updates a user in the local database after they sign in via Clerk. If the user already exists the name, email and avatar are refreshed.

Request URL: `http://localhost:3001/api/auth/sync`

Request:

```
{
  "clerkId": "user_2abc123xyz",
  "name": "Mehedi Hasan",
  "email": "mehedi@example.com",
  "imageUrl": "https://img.clerk.com/avatar.jpg"
}
```

Response:

```
{
  "success": true,
  "user": {
    "_id": "user_2abc123xyz",
    "name": "Mehedi Hasan",
    "email": "mehedi@example.com",
    "imageUrl": "https://img.clerk.com/avatar.jpg",
    "cartItems": {},
    "wishlist": [],
    "createdAt": "2025-01-15T08:00:00.000Z",
    "updatedAt": "2025-01-15T08:00:00.000Z"
  }
}
```

User Authentication & Profile

Get user profile

GET

/api/users/profile

AUTH

Returns the profile of the currently authenticated user.

Request URL: `http://localhost:3001/api/users/profile`

Response:

```
{
  "_id": "user_381WfIjro5DSC01FjSH1VSSAUTH",
  "name": "Md. Mehedi Hasan",
  "email": "mehedi@example.com",
  "imageUrl": "https://img.clerk.com/temp",
  "cartItemCount": 0,
  "wishlistCount": 0
}
```

User Authentication & Profile

Update user profile

PUT

/api/user/profile

AUTH

Updates the name and/or avatar URL of the authenticated user.

Request URL: `http://localhost:3001/api/users/profile`

Request:

```
{
  "name": "string",
  "imageUrl": "string"
}
```

Response:

```
{
  "success": true,
  "user": {
    "_id": "user_381WfIjro5DSC01FjSH1VSSAUTH",
    "name": "string",
    "email": "mehedi@example.com",
    "imageUrl": "string",
    "__v": 1,
    "wishlist": [],
    "updatedAt": "2026-02-28T15:34:25.689Z"
  }
}
```

PRODUCTS

Product Catalog Endpoints

GET	/api/products	PUBLIC
	List products with pagination, filter and sorting	
GET	/api/products/search	PUBLIC
	Search products by title, category name or ASIN	
GET	/api/products/categories	PUBLIC
	Get all categories with product counts	
GET	/api/products/best-sellers	PUBLIC
	Get best-selling products by popularity	
GET	/api/products/:id	PUBLIC
	Get a single product by its ID	
GET	/api/products/:id/related	PUBLIC
	Get related products in the same category	

PRODUCTS

Product Catalog Endpoints

POST

/api/products

Add new product

AUTH

PUT

/api/products/:id

Update specific product

AUTH

DELETE

/api/products/:id

Remove specific product

AUTH

Products

List products with pagination, filter and sorting

GET

/api/products

PUBLIC

Returns a paginated list of products. Supports filtering by category, price range, minimum rating, best-seller flag, and sorting by price / stars / reviews / popularity.

Request URL:

```
http://localhost:3001/api/products  
?page=1&limit=1&sortBy=price&sortOrder=asc&category=Men%27s%20Shoes
```

Products

List Products with pagination, filters, and sorting

GET

/api/products

PUBLIC

Request Parameters:

Parameter	Type	Description
page	integer	Page number. Default: 1
limit	integer	Items per page, max 100. Default: 20
sortBy	string	price stars reviews boughtInLastMonth
sortOrder	string	asc desc. Default: desc
category	string	Category name (case-insensitive partial match)
minPrice	number	Minimum price
maxPrice	number	Maximum price
minRating	number	Minimum star rating (0-5)
bestSeller	string	"true" to return best sellers only

Response:

```
{
  "products": [
    {
      "_id": "6989a8237a63ab7035a9e955",
      "asin": "B08B3J9NLX",
      "title": "Men's Romaleo 4 Cross Trainer Shoes",
      "imgUrl": "https://m.media-amazon.com/images/I/71wpQC2WeFL._AC_UL320_.jpg",
      "productURL": "https://www.amazon.com/dp/B08B3J9NLX",
      "stars": 4.1,
      "reviews": 0,
      "price": 0,
      "listPrice": 0,
      "category_id": 114,
      "isBestSeller": false,
      "boughtInLastMonth": 0,
      "id": 114,
      "category_name": "Men's Shoes"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 1,
    "total": 249,
    "totalPages": 249,
    "hasNext": true,
    "hasPrev": false
  }
}
```

Products

Search products by title, category name or ASIN

GET `/api/products/search` PUBLIC

Performs a case-insensitive search across product title, category name, and ASIN.

Request URL:

```
http://localhost:3001/api/products/search  
?query=wireless%20headphones&page=1&limit=1
```

Products

Search Products by Title, Category, or ASIN

GET

/api/products/search

PUBLIC

Request Parameters:

Parameter	Type	Description
query	string	Required. Search keyword(s)
page	integer	Page number. Default: 1
limit	integer	Items per page, max 100. Default: 20

Response:

```
{
  "products": [
    {
      "_id": "6989a8237a63ab7035a9ec07",
      "asin": "B0C5MS74JR",
      "title": "Wireless Earbuds, 60H Playback Bluetooth 5.3 Headphones,Noise Cancelling Wireless Headphones with LED Battery Display, 4 Mics Clear Call, IPX7 Waterproof Bluetooth Earbuds for Workout Sports",
      "imgUrl": "https://m.media-amazon.com/sample.jpg",
      "productURL": "https://www.amazon.com/dp/B0C5MS74JR",

```

```
      "stars": 4.8,
      "reviews": 0,
      "price": 19.99,
      "listPrice": 29.99,
      "category_id": 75,
      "isBestSeller": false,
      "boughtInLastMonth": 0,
      "id": 75,
      "category_name": "Cell Phones & Accessories"
    }
  ],

```

```
  "pagination": {
    "page": 1,
    "limit": 1,
    "total": 11,
    "totalPages": 11,
    "hasNext": true,
    "hasPrev": false
  }
}
```

Products

Get all categories with product counts

GET

/api/products/categories

PUBLIC

Returns all product categories sorted by product count (descending).

Request URL: `http://localhost:3001/api/products/categories`

Response:

```
[
  {
    "count": 201, "categoryId": 91, "name": "Girls' Clothing"
  },
  {
    "count": 173, "categoryId": 84, "name": "Boys' Clothing"
  },
  {
    "count": 146, "categoryId": 270, "name": "Toys & Games"
  },
  {
    "count": 139, "categoryId": 114, "name": "Men's Shoes"
  }
]
```

Products

Get best-selling products by popularity

GET

/api/products/best-sellers

PUBLIC

Returns top best-selling products sorted by units bought in the last month.

Request URL: `http://localhost:3001/api/products/best-sellers?limit=2`

Response:

```
[
  {
    "_id": "6989a8247a63ab7035aa0a74",
    "asin": "B07QR4Q42L",
    "title": "Portable Air Compressor Air Pump",
    "imgUrl": "https://m.media-amazon.com/sample.jpg",
    "productURL": "https://www.amazon.com/dp/B07QR4Q42L",
    "stars": 4.5,
    "reviews": 73714,
    "price": 31.99,
    "listPrice": 44.99,
    "category_id": 16,
    "isBestSeller": true,
    "boughtInLastMonth": 20000,
    "id": 16,
    "category_name": "Automotive Tires & Wheels"
  },
  {
    "_id": "6989a8247a63ab7035aa0a75",
    "asin": "B07QR4Q42M",
    "title": "Portable Air Compressor Air Pump",
    "imgUrl": "https://m.media-amazon.com/sample.jpg",
    "productURL": "https://www.amazon.com/dp/B07QR4Q42M",
    "stars": 4.5,
    "reviews": 73714,
    "price": 31.99,
    "listPrice": 44.99,
    "category_id": 16,
    "isBestSeller": true,
    "boughtInLastMonth": 20000,
    "id": 16,
    "category_name": "Automotive Tires & Wheels"
  }
]
```

```
{
  "_id": "6989a8247a63ab7035a9f951",
  "asin": "B0BBQX7P5J",
  "title": "Portable Sound Machine",
  "imgUrl": "https://m.media-amazon.com/sample2.jpg",
  "productURL": "https://www.amazon.com/dp/B0BBQX7P5J",
  "stars": 4.7,
  "reviews": 2626,
  "price": 20.99,
  "listPrice": 0,
  "category_id": 128,
  "isBestSeller": true,
  "boughtInLastMonth": 10000,
  "id": 128,
  "category_name": "Wellness & Relaxation Products"
}
```

Products

Get a single product

GET

/api/products/:id

PUBLIC

Request URL: `http://localhost:3001/api/products/6989a8247a63ab7035a9f951`

Response:

```
{
  "_id": "6989a8247a63ab7035a9f951",
  "asin": "B0BBQX7P5J",
  "title": "Portable Sound Machine",
  "imgUrl": "https://m.media-amazon.com/sample2.jpg",
  "productURL": "https://www.amazon.com/dp/B0BBQX7P5J",
  "stars": 4.7,
  "reviews": 2626,
  "price": 20.99,
  "listPrice": 0,
  "category_id": 128,
  "isBestSeller": true,
  "boughtInLastMonth": 10000,
  "id": 128,
  "category_name": "Wellness & Relaxation Products"
}
```

Products

Get related products

GET

/api/products/:id/related

PUBLIC

Request Parameters:

Parameter	Type	Description
limit	integer	Items per page, max 100. Default: 20

Request URL: `http://localhost:3001/api/products/6989a8247a63ab7035aa0a74/related?limit=2`

Response:

```
[
  {
    "_id": "6989a8247a63ab7035aa0856",
    "asin": "B08Z78VVV6",
    "title": "Carwiner Jack Pad",
    "imgUrl": "https://m.media-amazon.com/sample3.jpg",
    "productURL": "https://www.amazon.com/dp/B08Z78VVV6",
    "stars": 4.7,
    "reviews": 750,
    "price": 17.99,
    "listPrice": 0,
    "category_id": 16,
    "isBestSeller": false,
    "boughtInLastMonth": 700,
    "id": 16,
    "category_name": "Automotive Tires & Wheels"
  },
]
```

```
{
  "_id": "6989a8247a63ab7035a9fcf9",
  "asin": "B0873WXTPV",
  "title": "TPMS Sensor, 4 Pack Tire Sensor",
  "imgUrl": "https://m.media-amazon.com/sample4.jpg",
  "productURL": "https://www.amazon.com/dp/B0873WXTPV",
  "stars": 4.1,
  "reviews": 95,
  "price": 32.99,
  "listPrice": 0,
  "category_id": 16,
  "isBestSeller": false,
  "boughtInLastMonth": 600,
  "id": 16,
  "category_name": "Automotive Tires & Wheels"
}
```

Products

Add new product

POST

/api/products

AUTH

Request URL: `http://localhost:3001/api/products`

Request:

```
{
  "categoryId": "electronics-001",
  "name": "Wireless Bluetooth Headphones",
  "description": "High-quality wireless headphones with noise cancellation and 30-hour battery life",
  "price": 79.99,
  "stockQty": 150
}
```

Response:

```
{
  "success": true,
  "message": "Product created successfully",
  "data": {
    "productId": "prod_abc123xyz",
    "categoryId": "electronics-001",
    "name": "Wireless Bluetooth Headphones",
    "description": "High-quality wireless headphones with noise cancellation and 30-hour battery life",
    "price": 79.99,
    "rating": 0.0,
    "stockQty": 150,
    "createdAt": "2025-02-21T10:30:00Z"
  }
}
```

Products

Update specific product

PUT

/api/products/:id

AUTH

Request URL: `http://localhost:3001/api/products/prod_abc123xyz`

Request:

```
{
  "categoryId": "electronics-001",
  "name": "Wireless Bluetooth Headphones",
  "description": "High-quality wireless headphones with noise cancellation and 30-hour battery life",
  "price": 79.99,
  "stockQty": 150
}
```

Response:

```
{
  "success": true,
  "message": "Product updated successfully",
  "data": {
    "productId": "prod_abc123xyz",
    "categoryId": "electronics-001",
    "name": "Wireless Bluetooth Headphones",
    "description": "High-quality wireless headphones with noise cancellation and 30-hour battery life",
    "price": 79.99,
    "rating": 0.0,
    "stockQty": 150,
    "createdAt": "2025-02-21T10:30:00Z"
  }
}
```

Products

Remove specific product

DELETE

/api/products/:id

AUTH

Request URL: `http://localhost:3001/api/products/prod_abc123xyz`

Response:

```
{  
  "success": true  
}
```

CART

Shopping Cart Endpoints

GET

`/api/cart`

AUTH

Get cart items with product details and totals

POST

`/api/cart/update`

AUTH

Add item, update quantity, or remove item

DELETE

`/api/cart/clear`

AUTH

Remove all items from the cart

CART

Get cart items with product details and totals

GET

/api/cart

AUTH

Returns the authenticated user's cart with full product details, per-item subtotals, total price, and total item count

Request URL: `http://localhost:3001/api/cart`

CART

Get cart items with product details and totals

GET

/api/cart

AUTH

Response:

```
{
  "items": [
    {
      "product": {
        "_id": "6989a8247a63ab7035aa0a74",
        "asin": "B07QR4Q42L",
        "title": "Portable Air Compressor Air Pump",
        "imgUrl": "https://m.media-amazon.com/sample.jpg",
        "productURL": "https://www.amazon.com/dp/B07QR4Q42L",
        "stars": 4.5,
        "reviews": 73714,
        "price": 31.99,
        "listPrice": 44.99,
        "category_id": 16,
        "isBestSeller": true,
        "boughtInLastMonth": 20000,
        "id": 16,
        "category_name": "Automotive Tires & Wheels"
      },
      "quantity": 1,
      "subtotal": 31.99
    }
  ],
  "total": 31.99,
  "itemCount": 1
}
```

CART

Add item, update quantity, or remove item from cart

POST

/api/cart/update

AUTH

Request URL: `http://localhost:3001/api/cart/update`

Request Body:

```
{
  "productId": "6989a8247a63ab7035aa0a74",
  "quantity": 1
}
```

Response:

```
[
  "items": [
    {
      "product": {
        "_id": "6989a8247a63ab7035aa0a74",
        "asin": "B07QR4Q42L",
        "title": "Portable Air Compressor Air Pump",
        "imgUrl": "https://m.media-amazon.com/sample.jpg",
        "productURL": "https://www.amazon.com/dp/B07QR4Q42L",
        "stars": 4.5,
        "reviews": 73714,
        "price": 31.99,
        "listPrice": 44.99,
        "category_id": 16,
        "isBestSeller": true,
        "boughtInLastMonth": 20000,
        "id": 16,
        "category_name": "Automotive Tires & Wheels"
      },
      "quantity": 1,
      "subtotal": 31.99
    }
  ],
  "total": 31.99,
  "itemCount": 1
}
```

CART

Remove all items from the cart

DELETE

/api/cart/clear

AUTH

Request URL: `http://localhost:3001/api/cart/clear`

Response:

```
{  
  "items": [],  
  "total": 0,  
  "itemCount": 0  
}
```

ORDERS

Order Management Endpoints

POST	/api/orders Create new order	AUTH
GET	/api/orders Get current user's orders with pagination	AUTH
GET	/api/orders/:id Get full order details with status history	AUTH
POST	/api/orders/:id/cancel Cancel a pending or confirmed order	AUTH
GET	/api/orders/all Get all the orders in the system	AUTH
GET	/api/orders/stats Get order count and revenue statistics	AUTH
POST	/api/orders/:id/update-status Advance order through the status lifecycle	AUTH

ORDERS

Create new order

POST

/api/orders

AUTH

Request URL: `http://localhost:3001/api/orders`

Request:

```
{
  "items": [
    {
      "product": "6989a8247a63ab7035aa0a74",
      "quantity": 1
    }
  ],
  "addressId": "69611987679e16c21655a551",
  "paymentMethod": "COD"
}
```

Response:

```
{
  "success": true,
  "order": {
    "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
    "items": [
      {
        "product": "6989a8247a63ab7035aa0a74",
        "quantity": 1,
        "price": 31.99
      }
    ],
    "amount": 32.63,
    "address": "69611987679e16c21655a551",
    "status": "Pending",
    "paymentMethod": "COD",
    "paymentStatus": "Pending",
    "paymentId": null,
    "paidAt": null,
    "trackingNumber": null,
    "estimatedDelivery": null,

```

```
    "statusHistory": [
      {
        "status": "Pending",
        "timestamp": "2026-02-28T16:41:51.432Z",
        "note": "Order placed"
      }
    ],
    "_id": "69a31acfb8d7e45c57148ec5",
    "createdAt": "2026-02-28T16:41:51.448Z",
    "updatedAt": "2026-02-28T16:41:51.448Z",
    "__v": 0
  }
}
```

ORDERS

Get current user's orders with pagination

GET

/api/orders

AUTH

Request URL: `http://localhost:3001/api/orders?page=1&limit=1`

Response:

```
{
  "orders": [
    {
      "_id": "69a31acfb8d7e45c57148ec5",
      "userId": "user_381WfIjro5DSCO1FjSH1VSSAUTH",
      "items": [
        {
          "product": {
            "_id": "6989a8247a63ab7035aa0a74",
            "asin": "B07QR4Q42L",
            "title": "Portable Air Compressor Air Pump",
            "imgUrl": "https://m.media-amazon.com/sample.jpg",
            "productURL": "https://www.amazon.com/dp/B07QR4Q42L",
            "stars": 4.5,
            "reviews": 73714,
            "price": 31.99,
            "listPrice": 44.99,
            "category_id": 16,
            "isBestSeller": true,
            "boughtInLastMonth": 20000,
            "id": 16,
            "category_name": "Automotive Tires & Wheels"
          },
          "quantity": 1,
          "price": 31.99
        }
      ],
      "amount": 32.63,
      "address": {
        "_id": "69611987679e16c21655a551",
        "userId": "user_381WfIjro5DSCO1FjSH1VSSAUTH",
        "fullName": "Md. Mehedi Hasan",
        "phoneNumber": "01234567890",
        "pincode": 1000,
        "area": "Palashi",
        "city": "Dhaka",
        "state": "Dhaka",
        "__v": 0
      },
      "status": "Pending",
      "paymentMethod": "COD",
      "paymentStatus": "Pending",
      "paymentId": null,
      "paidAt": null,
      "trackingNumber": null,
      "estimatedDelivery": null,
      "statusHistory": [
        {
          "status": "Pending",
          "timestamp": "2026-02-28T16:41:51.432Z",
          "note": "Order placed"
        }
      ]
    },
    {
      "createdAt": "2026-02-28T16:41:51.448Z",
      "updatedAt": "2026-02-28T16:41:51.448Z",
      "__v": 0
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 1,
    "total": 15,
    "totalPages": 15
  }
}
```

ORDERS

Get full order details with status history

GET

/api/orders/:id

AUTH

Request URL: `http://localhost:3001/api/orders/69a31acfb8d7e45c57148ec5`

Response:

```
{
  "_id": "69a31acfb8d7e45c57148ec5",
  "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
  "items": [
    {
      "product": {
        "_id": "6989a8247a63ab7035aa0a74",
        "asin": "B07QR4Q42L",
        "title": "Portable Air Compressor Air Pump",
        "imgUrl": "https://m.media-amazon.com/sample.jpg",
        "productURL": "https://www.amazon.com/dp/B07QR4Q42L",
        "stars": 4.5,
        "reviews": 73714,
        "price": 31.99,
        "listPrice": 44.99,
        "category_id": 16,
        "isBestSeller": true,
        "boughtInLastMonth": 20000,
        "id": 16,
        "category_name": "Automotive Tires & Wheels"
      },
      "quantity": 1,
      "price": 31.99
    }
  ]
}
```

```
"amount": 32.63,
"address": {
  "_id": "69611987679e16c21655a551",
  "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
  "fullName": "Md. Mehedi Hasan",
  "phoneNumber": "01234567890",
  "pincode": 1000,
  "area": "Palashi",
  "city": "Dhaka",
  "state": "Dhaka",
  "__v": 0
},
"status": "Pending",
"paymentMethod": "COD",
"paymentStatus": "Pending",
"paymentId": null,
"paidAt": null,
"trackingNumber": null,
"estimatedDelivery": null,
"statusHistory": [
  {
    "status": "Pending",
    "timestamp": "2026-02-28T16:41:51.432Z",
    "note": "Order placed"
  }
],
"createdAt": "2026-02-28T16:41:51.448Z",
"updatedAt": "2026-02-28T16:41:51.448Z",
"__v": 0
}
```

ORDERS

Cancel a pending or confirmed order

POST

/api/orders/:id/cancel

AUTH

Request URL: `http://localhost:3001/api/orders/69a31acfb8d7e45c57148ec5/cancel`

Response:

```
{
  "success": true,
  "order": {
    "_id": "69a31acfb8d7e45c57148ec5",
    "userId": "user_381WfIjro5DSCO1FjSH1VSSAUth",
    "items": [
      {
        "product": "6989a8247a63ab7035aa0a74",
        "quantity": 1,
        "price": 31.99
      }
    ],
    "amount": 32.63,
    "address": "69611987679e16c21655a551",
    "status": "Cancelled",
    "paymentMethod": "COD",
    "paymentStatus": "Pending",
    "paymentId": null,
    "paidAt": null,
    "trackingNumber": null,
    "estimatedDelivery": null,
  }
}
```

```
"statusHistory": [
  {
    "status": "Pending",
    "timestamp": "2026-02-28T16:41:51.432Z",
    "note": "Order placed"
  },
  {
    "status": "Confirmed",
    "timestamp": "2026-02-28T17:07:21.301Z",
    "note": "Sample Note"
  },
  {
    "status": "Cancelled",
    "timestamp": "2026-02-28T17:11:32.342Z",
    "note": "Cancelled by customer"
  }
],
"createdAt": "2026-02-28T16:41:51.448Z",
"updatedAt": "2026-02-28T17:11:32.347Z",
"__v": 2
}
```

ORDERS

Get all the orders in the system

GET

/api/orders/all

AUTH

Request URL: `http://localhost:3001/api/orders/all?page=1&limit=1`

Response:

```
{
  "orders": [
    {
      "_id": "69a31acfb8d7e45c57148ec5",
      "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
      "items": [
        {
          "product": {
            "_id": "6989a8247a63ab7035aa0a74",
            "asin": "B07QR4Q42L",
            "title": "Portable Air Compressor Air Pump",
            "imgUrl": "https://m.media-amazon.com/sample.jpg",
            "productURL": "https://www.amazon.com/dp/B07QR4Q42L",
            "stars": 4.5,
            "reviews": 73714,
            "price": 31.99,
            "listPrice": 44.99,
            "category_id": 16,
            "isBestSeller": true,
            "boughtInLastMonth": 20000,
            "id": 16,
            "category_name": "Automotive Tires & Wheels"
          },
          "quantity": 1,
          "price": 31.99
        }
      ]
    }
  ]
}
```

```
    "amount": 32.63,
    "address": {
      "_id": "69611987679e16c21655a551",
      "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
      "fullName": "Md. Mehedi Hasan",
      "phoneNumber": "01234567890",
      "pincode": 1000,
      "area": "Palashi",
      "city": "Dhaka",
      "state": "Dhaka",
      "__v": 0
    },
    "status": "Pending",
    "paymentMethod": "COD",
    "paymentStatus": "Pending",
    "paymentId": null,
    "paidAt": null,
    "trackingNumber": null,
    "estimatedDelivery": null,
    "statusHistory": [
      {
        "status": "Pending",
        "timestamp": "2026-02-28T16:41:51.432Z",
        "note": "Order placed"
      }
    ]
  ]
}
```

```
    "createdAt": "2026-02-28T16:41:51.448Z",
    "updatedAt": "2026-02-28T16:41:51.448Z",
    "__v": 0
  },
  "pagination": {
    "page": 1,
    "limit": 1,
    "total": 19,
    "totalPages": 19
  }
}
```

ORDERS

Get order count and revenue statistics

GET

/api/orders/stats

AUTH

Request URL: `http://localhost:3001/api/orders/stats`

Response:

```
{
  "statusBreakdown": [
    {
      "_id": "Order Placed", "count": 11, "totalAmount": 342543
    },
    {
      "_id": "Cancelled", "count": 2, "totalAmount": 67471
    },
    {
      "_id": "Processing", "count": 1, "totalAmount": 79560
    },
    {
      "_id": "Pending", "count": 4, "totalAmount": 1516.65
    },
    {
      "_id": "Confirmed", "count": 1, "totalAmount": 32.63
    }
  ],
  "totalOrders": 19,
  "totalRevenue": 79894.63
}
```

ORDERS

Advance order through the status lifecycle

POST

/api/orders/:id/update-status

AUTH

Request URL: `http://localhost:3001/api/orders/69a31acfb8d7e45c57148ec5/update-status`

Request:

```
{
  "status": "Confirmed",
  "note": "Sample Note"
}
```

Response:

```
{
  "success": true,
  "order": {
    "_id": "69a31acfb8d7e45c57148ec5",
    "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
    "items": [
      {
        "product": "6989a8247a63ab7035aa0a74",
        "quantity": 1,
        "price": 31.99
      }
    ],
    "amount": 32.63,
    "address": "69611987679e16c21655a551",
    "status": "Confirmed",
    "paymentMethod": "COD",
    "paymentStatus": "Pending",
    "paymentId": null,
    "paidAt": null,
    "trackingNumber": null,
    "estimatedDelivery": null,
  }
}
```

```
"statusHistory": [
  {
    "status": "Pending",
    "timestamp": "2026-02-28T16:41:51.432Z",
    "note": "Order placed"
  },
  {
    "status": "Confirmed",
    "timestamp": "2026-02-28T17:07:21.301Z",
    "note": "Sample Note"
  }
],
"createdAt": "2026-02-28T16:41:51.448Z",
"updatedAt": "2026-02-28T17:07:21.309Z",
"__v": 1
}
```

SHIPPING ADDRESS MANAGEMENT

Shipping Address management Endpoints

POST

/api/addresses

AUTH

Create a new shipping address

GET

/api/addresses

AUTH

Get all addresses for the current user

GET

/api/addresses/:id

AUTH

Get address by id

PUT

/api/addresses/:id

AUTH

Update an address

DELETE

/api/addresses/:id

AUTH

Delete an address

SHIPPING ADDRESS MANAGEMENT

Create a new shipping address

POST

/api/addresses

AUTH

Request URL: `http://localhost:3001/api/addresses`

Request:

```
{
  "fullName": "Sample Address",
  "phoneNumber": "01233456789",
  "pincode": 123456,
  "area": "Sample Area 1",
  "city": "Sample City 1",
  "state": "Sample State 1",
  "isDefault": true
}
```

Response:

```
{
  "success": true,
  "address": {
    "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
    "fullName": "Sample Address",
    "phoneNumber": "01233456789",
    "pincode": 123456,
    "area": "Sample Area 1",
    "city": "Sample City 1",
    "state": "Sample State 1",
    "isDefault": true,
    "_id": "69a322afb8d7e45c57148edc",
    "createdAt": "2026-02-28T17:15:27.305Z",
    "updatedAt": "2026-02-28T17:15:27.305Z",
    "__v": 0
  }
}
```

SHIPPING ADDRESS MANAGEMENT

Get all addresses for the current user

GET

/api/addresses

AUTH

Request URL: `http://localhost:3001/api/addresses`

Response:

```
[
  {
    "_id": "69a322afb8d7e45c57148edc",
    "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
    "fullName": "Sample Address",
    "phoneNumber": "01233456789",
    "pincode": 123456,
    "area": "Sample Area 1",
    "city": "Sample City 1",
    "state": "Sample State 1",
    "isDefault": true,
    "createdAt": "2026-02-28T17:15:27.305Z",
    "updatedAt": "2026-02-28T17:15:27.305Z",
    "__v": 0
  },
  {
    "_id": "69611987679e16c21655a551",
    "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
    "fullName": "Md. Mehedi Hasan",
    "phoneNumber": "01634681690",
    "pincode": 1000,
    "area": "Palashi",
    "city": "Dhaka",
    "state": "Dhaka",
    "__v": 0,
    "isDefault": false,
    "updatedAt": "2026-02-28T17:15:27.200Z"
  }
]
```

SHIPPING ADDRESS MANAGEMENT

Get address by id

GET

/api/addresses/:id

AUTH

Request URL: `http://localhost:3001/api/addresses/69a322afb8d7e45c57148edc`

Response:

```
{
  "_id": "69a322afb8d7e45c57148edc",
  "userId": "user_381WFIjro5DSCO1FjSH1VSSAuth",
  "fullName": "Sample Address",
  "phoneNumber": "01233456789",
  "pincode": 123456,
  "area": "Sample Area 1",
  "city": "Sample City 1",
  "state": "Sample State 1",
  "isDefault": true,
  "createdAt": "2026-02-28T17:15:27.305Z",
  "updatedAt": "2026-02-28T17:15:27.305Z",
  "__v": 0
}
```

SHIPPING ADDRESS MANAGEMENT

Update an address

PUT

/api/addresses/:id

AUTH

Request URL: `http://localhost:3001/api/addresses/69a322afb8d7e45c57148edc`

Request:

```
{
  "fullName": "Sample Address 2",
  "phoneNumber": "01233499999",
  "pincode": 123422,
  "area": "Sample Area 2",
  "city": "Sample City 2",
  "state": "Sample State 2",
  "isDefault": true
}
```

Response:

```
{
  "success": true,
  "address": {
    "_id": "69a322afb8d7e45c57148edc",
    "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
    "fullName": "Sample Address 2",
    "phoneNumber": "01233499999",
    "pincode": 123422,
    "area": "Sample Area 2",
    "city": "Sample City 2",
    "state": "Sample State 2",
    "isDefault": true,
    "createdAt": "2026-02-28T17:15:27.305Z",
    "updatedAt": "2026-02-28T17:22:11.448Z",
    "__v": 0
  }
}
```

SHIPPING ADDRESS MANAGEMENT

Delete an address

DELETE

/api/addresses/:id

AUTH

Request URL: `http://localhost:3001/api/addresses/69a322afb8d7e45c57148edc`

Response:

```
{  
  "success": true  
}
```

PRODUCT REVIEWS & RATINGS

Product Reviews & Ratings Endpoints

POST	/api/reviews Create a product review	AUTH
GET	/api/reviews/product/:productId Get reviews for a product	PUBLIC
GET	/api/reviews/my-reviews Get current user reviews	AUTH
PUT	/api/reviews/:id Update a review	AUTH
DELETE	/api/reviews/:id Delete a review	AUTH

PRODUCT REVIEWS & RATINGS

Create a product review

POST

/api/reviews

AUTH

Request URL: `http://localhost:3001/api/reviews`

Request:

```
{
  "productId": "6989a8247a63ab7035aa0a74",
  "rating": 5,
  "title": "Product Review",
  "comment": "This is a very helpful product."
}
```

Response:

```
{
  "success": true,
  "review": {
    "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
    "productId": "6989a8247a63ab7035aa0a74",
    "orderId": null,
    "rating": 5,
    "title": "Product Review",
    "comment": "This is a very helpful product.",
    "isVerifiedPurchase": false,
    "_id": "69a32763b8d7e45c57148ef0",
    "createdAt": "2026-02-28T17:35:31.252Z",
    "updatedAt": "2026-02-28T17:35:31.252Z",
    "__v": 0
  }
}
```

PRODUCT REVIEWS & RATINGS

Get reviews for a product

GET

/api/reviews/product/:productId

PUBLIC

Request URL: `http://localhost:3001/api/reviews/product/6989a8247a63ab7035aa0a74?page=1&limit=1`

Response:

```
{
  "reviews": [
    {
      "_id": "69a32763b8d7e45c57148ef0",
      "userId": "user_381WfIjro5DSC01FjSH1VSSAUth",
      "productId": "6989a8247a63ab7035aa0a74",
      "orderId": null,
      "rating": 5,
      "title": "Product Review",
      "comment": "This is a very helpful product.",
      "isVerifiedPurchase": false,
      "createdAt": "2026-02-28T17:35:31.252Z",
      "updatedAt": "2026-02-28T17:35:31.252Z",
      "__v": 0,
      "userName": "string",
      "userImage": "string"
    }
  ],
}
```

```
  "pagination": {
    "page": 1,
    "limit": 1,
    "total": 1,
    "totalPages": 1
  },
  "summary": {
    "averageRating": 5,
    "totalReviews": 1,
    "ratingDistribution": {
      "1": 0,
      "2": 0,
      "3": 0,
      "4": 0,
      "5": 1
    }
  }
}
```

PRODUCT REVIEWS & RATINGS

Get current user reviews

GET

/api/reviews/my-reviews

AUTH

Request URL: `http://localhost:3001/api/reviews/my-reviews`

Response:

```
{
  "_id": "69a32763b8d7e45c57148ef0",
  "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
  "productId": {
    "_id": "6989a8247a63ab7035aa0a74",
    "asin": "B07QR4Q42L",
    "title": "Portable Air Compressor Air Pump",
    "imgUrl": "https://m.media-amazon.com/sample.jpg",
    "productURL": "https://www.amazon.com/dp/B07QR4Q42L",
    "stars": 4.5,
    "reviews": 73714,
    "price": 31.99,
    "listPrice": 44.99,
    "category_id": 16,
    "isBestSeller": true,
    "boughtInLastMonth": 20000,
    "id": 16,
    "category_name": "Automotive Tires & Wheels"
  },
  "orderId": null,
  "rating": 5,
  "title": "Product Review",
  "comment": "This is a very helpful product.",
  "isVerifiedPurchase": false,
  "createdAt": "2026-02-28T17:35:31.252Z",
  "updatedAt": "2026-02-28T17:35:31.252Z",
  "__v": 0
},
```

```
{
  "_id": "69a04e73e012dc8dec77c824",
  "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
  "productId": {
    "_id": "6989a8237a63ab7035a9e88b",
    "asin": "B075QKLD5",
    "title": "Battery 11.1V 2330mAh",
    "imgUrl": "https://m.media-amazon.com/sample2.jpg",
    "productURL": "https://www.amazon.com/dp/B075QKLD5",
    "stars": 4.3,
    "reviews": 0,
    "price": 26.99,
    "listPrice": 0,
    "category_id": 65,
    "isBestSeller": false,
    "boughtInLastMonth": 0,
    "id": 65,
    "category_name": "Laptop Accessories"
  },
  "orderId": null,
  "rating": 5,
  "title": "",
  "comment": "",
  "isVerifiedPurchase": false,
  "createdAt": "2026-02-26T13:45:23.388Z",
  "updatedAt": "2026-02-26T13:45:23.388Z",
  "__v": 0
}
```

PRODUCT REVIEWS & RATINGS

Update a review

PUT

/api/reviews/:id

AUTH

Request URL: `http://localhost:3001/api/reviews/69a32763b8d7e45c57148ef0`

Request:

```
{
  "rating": 5,
  "title": "Product Review v2",
  "comment": "This is a very helpful product v2"
}
```

Response:

```
{
  "success": true,
  "review": {
    "_id": "69a32763b8d7e45c57148ef0",
    "userId": "user_381WfIjro5DSC01FjSH1VSSAUTH",
    "productId": "6989a8247a63ab7035aa0a74",
    "orderId": null,
    "rating": 5,
    "title": "Product Review v2",
    "comment": "This is a very helpful product v2",
    "isVerifiedPurchase": false,
    "createdAt": "2026-02-28T17:35:31.252Z",
    "updatedAt": "2026-02-28T17:49:07.134Z",
    "__v": 0
  }
}
```

PRODUCT REVIEWS & RATINGS

Delete a review

DELETE

/api/reviews/:id

AUTH

Request URL: `http://localhost:3001/api/reviews/69a32763b8d7e45c57148ef0`

Response:

```
{  
  "success": true  
}
```

USER WISHLIST MANAGEMENT

User Wishlist Endpoints

GET

/api/wishlist

Get wishlist items

AUTH

POST

/api/wishlist/:productId

Add product to wishlist

AUTH

DELETE

/api/wishlist/:productId

Remove product from wishlist

AUTH

GET

/api/wishlist/:productId/check

Check if the product is in wishlist

AUTH

USER WISHLIST MANAGEMENT

Get wishlist items

GET

/api/wishlist

AUTH

Request URL: `http://localhost:3001/api/wishlist`

Response:

```
{
  "items": [
    {
      "_id": "6989a8247a63ab7035aa0a74",
      "asin": "B07QR4Q42L",
      "title": "Portable Air Compressor Air Pump",
      "imgUrl": "https://m.media-amazon.com/sample.jpg",
      "productURL": "https://www.amazon.com/dp/B07QR4Q42L",
      "stars": 4.5,
      "reviews": 73714,
      "price": 31.99,
      "listPrice": 44.99,
      "category_id": 16,
      "isBestSeller": true,
      "boughtInLastMonth": 20000,
      "id": 16,
      "category_name": "Automotive Tires & Wheels"
    }
  ]
}
```

USER WISHLIST MANAGEMENT

Add product to wishlist

POST

/api/wishlist/:productId

AUTH

Request URL: `http://localhost:3001/api/wishlist/6989a8247a63ab7035aa0a74`

Response:

```
{  
  "success": true,  
  "wishlistCount": 1  
}
```

USER WISHLIST MANAGEMENT

Remove product from wishlist

DELETE

/api/wishlist/:productId

AUTH

Request URL: `http://localhost:3001/api/wishlist/6989a8247a63ab7035aa0a74`

Response:

```
{
  "success": true,
  "wishlistCount": 0
}
```

USER WISHLIST MANAGEMENT

Check if the product is in wishlist

GET

/api/wishlist/:productId/check

AUTH

Request URL: `http://localhost:3001/api/wishlist/6989a8247a63ab7035aa0a74/check`

Response:

```
{  
  "inWishlist": true  
}
```

STRIPE PAYMENT INTEGRATION

Stripe payment integration Endpoints

POST

`/api/payments/checkout/:orderId`

AUTH

Create Stripe checkout session for an order

POST

`/api/payments/verify`

AUTH

Verify Stripe payment by session ID

STRIPE PAYMENT INTEGRATION

Create Stripe checkout session for an order

POST

/api/payments/checkout/:orderId

AUTH

Request URL: `http://localhost:3001/api/payments/checkout/69a32d09b8d7e45c57148f0a`

Response:

```
{
  "sessionId": "cs_test_b1qiadqCBGNqqbZX6g0LWhv0XL0hP6UHj1QKSMwYbZ3yvT8Pmcc5oAC2ok",
  "url": "https://checkout.stripe.com/c/pay/cs_test_b1qiadqCBGNqqbZX6g0LWhv0XL0hP6UHj1QKSMwYbZ3yvT8Pmcc5oAC2ok#fidnandhYHdWcXxpYCc%2FJ2FgY2RwaXEnKSdkdWxOYHwnPyd1blpxYHZxWjA0VmtvUnU1TEFycEFMSX9mQnE8T0NmPD1LXEBBZGBPYGpyXTxQVlJ3b1ddYWdnfH89bmhfNTM2U3JnSGd2dn1sUEpoRjRCcGRtbGpLSDI0STFMNXBiS3U0NTU2R2oyQ1NzPCcpJ2N3amhWYHdzYHcnP3F3cGApJ2dkZm5id2pwa2FGam1qdyc%2FJyZjY2NjY2MnKSdpZHxqcHFRfHVgJz8naHBpcWxabHFgaCcpJ2BrZGdpYFVpZGZgbWppYWb3dic%2FcXdwYHg1"
}
```

STRIPE PAYMENT INTEGRATION

Verify Stripe payment by session ID

POST

/api/payments/verify

AUTH

Request URL: `http://localhost:3001/api/payments/verify`

Response:

```
{
  "success": true,
  "paymentStatus": "Paid",
  "order": {
    "_id": "67a1b2c3d4e5f6789abcdef5",
    "status": "Confirmed",
    "paymentStatus": "Paid",
    "paymentId": "pi_3NxyzABC1234567890",
    "paidAt": "2025-02-01T10:15:00.000Z",
    "statusHistory": [
      {
        "status": "Pending",
        "timestamp": "2025-02-01T10:00:00.000Z"
      },
      {
        "status": "Confirmed",
        "timestamp": "2025-02-01T10:15:00.000Z",
        "note": "Payment received via Stripe"
      }
    ]
  }
}
```

THANK YOU